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Key factors in predictive analysis of cardiovascular risks in public health

Ghazi I. Al Jowf[1]✉ & Manjur Kolhar[2]

This research emphasizes the role of analytics in evaluating the risk of disease (CVD) focusing on thorough data preparation and feature engineering for accurate predictions. We studied machine learning (ML) and learning (DL) models, such as Logistic Regression (LR) Random Forest (RF) Gradient Boosting Machines (GBM) and Multilayer Perceptron (MLP). Each model's performance was assessed using metrics like accuracy, precision, recall, F1 score and ROC AUC to determine their reliability and practical relevance. Our analysis shows the strengths of each model category. Conventional ML models like Random Forest and Gradient Boosting Machines were effective in identifying patients at risk achieving up to 74% accuracy and 72% recall. On the hand, deep learning models like Multilayer Perceptron excelled in handling data with an impressive ROC AUC score of approximately 80%. Despite the need for resources and extensive data preprocessing these models are highly skilled at pinpointing crucial risk factors, crucial, for long term CVD management.

Keywords Cardiovascular, Machine learning, Deep learning, Regression, Random forest, Gradient boosting machines, Feature selection

Cardiovascular diseases are a key concern, because they are primary cause of death worldwide. Detecting and managing these diseases is crucial, and it remains challenging due to current limitations[1,2]. The interplay of demographic and lifestyle factors in heart health emphasizes the need for more advanced data driven methods. Early intervention has the potential to change the course of diseases, reducing mortality rates and enhancing quality of life. This sense of urgency is driving the exploration of machine learning (ML) technologies as a frontier. The ability of machine learning and deep learning to analyze datasets can uncover insights into cardiovascular risk factors leading to more precise and timely predictions. These advancements play a role in expanding healthcare strategies that proactively address cardiovascular disease risks. Our research contributes to these efforts by exploring how novel approaches in machine learning (ML) and deep learning (DL) techniques can enhance the ability to identify and control the early signs of cardiovascular risks. Considering the need for innovation in the detection and management of cardiovascular diseases, our study proposes a holistic approach to leveraging machine learning and DL features to address the multiple factors that are involved in the risk of cardiovascular disease.

Our objective is to capture the nonlinear connections among heart disease risk factors by introducing interaction terms and polynomial features. Additionally, we incorporate Recursive Feature Elimination (RFE) to enable feature selections, hence removing variables to streamline and strengthen our analysis. By combining machine learning techniques with feature engineering and careful feature selection we aim to enhance the accuracy of predicting heart disease risks and shed light on how technology can benefit healthcare. Through demonstrating the effectiveness of these approaches our hope is to pave the way for their use, in settings ultimately assisting in detection and proactive management of heart diseases.

The primary reason for our choosing Recursive Feature Elimination (RFE) was its efficiency in identifying clinically meaningful features that are directly linked to cardiovascular risk factors without compromising on the feature interpretability that is crucial for clinical decision making. Although methods like PCA reduce the dimensionality of the data quickly, they make the features uninterpretable and Mutual Information or LASSO select features that are statistically related but may not include clinically important predictors. However, we do acknowledge that expanding the search to other techniques may only strengthen the results of our study. This paper examines the potential of ML and DL to improve CVD prediction. We begin by surveying the existing research landscape in this field (literature survey). Next, we unveil our proposed methodology, detailing our data analysis approach, model selection strategies, and how we fine-tuned the data (feature engineering). The heart of the paper lies in the results section, where we showcase the predictive power of various models. Building on these findings, the discussion section explores their significance in the context of broader CVD risk assessment

[1]Department Public Health, College of Applied Medical Sciences, King Faisal University, Al Hofuf 37912, Al Ahsa, Saudi Arabia. [2]Department Health Information Management and Technology, College of Applied Medical Sciences, King Faisal University, Al Hofuf 37912, Al Ahsa, Saudi Arabia. ✉email: galjowf@kfu.edu.sa

and delves into potential clinical implications. Finally, we conclude by outlining exciting avenues for future research and emphasizing the potential for translating our findings into real-world clinical practice.

Literature review

Employing the Auto Prognosis ML framework on data from 423,604 UK Biobank participants, the authors demonstrated enhanced CVD risk prediction, achieving an AUC-ROC of 0.774 and outperforming traditional benchmarks. This work highlights unusual conjectures like walking pace and self-reported health and reams of other data integrated to enhance risk assessment[3]. In this reference[4], the ML procedures are applied to a UCI dataset of 303 records and 76 features, and 14 key structures are used to evaluate the performance of various methods in healthcare predictions. Naive Bayes, SVM, LR, Decision Trees, RF and KNN are compared. The research focuses on the use of KNN with eight neighbors because it has been found to give the best effectiveness, sensitivity, precision, accuracy and F1 score. This shows how it can effectively be used for disease prediction using healthcare data compared to other methods[4]. In[5] we introduce SCORE2 as a model to estimate the 10-year risk of nonfatal cardiovascular disease in people aged 40–69 years old without previous cardiovascular disease or diabetes from Europe. With data from 45 cohorts across 13 countries (677,684 people; 30,121 CVD incidents) the model includes variables such as age, blood pressure and cholesterol levels. It also adjusts for four risk regions by recalibrating based on cardiovascular disease mortality rates and distributions of risk factors. The external validation in 25 additional cohorts (1,133,181 individuals, 43,492 CVD events) showed C-indices between 0.67 and 0.81, indicating variable CVD risk across Europe. SCORE2 significantly advances the precision of 10-year CVD risk prediction, facilitating improved risk stratification across diverse European populations[5]. Authors[6], a CVD prediction model was developed from 29,930 high-risk individuals identified from a cohort of 101,056 in 2014, utilizing data from electronic health records. Logistic regression identified nearly 30 CVD-related indicators such as age, gender, lifestyle factors, and various health metrics. Multiple predictive methods including Random Forest, Ada Boost, Bagged Trees, Naïve Bayes, and CART were evaluated against a multivariate regression benchmark (AUC = 0.7143). Random Forest outperformed all with an AUC of 0.787, providing a robust 3-year CVD risk assessment tool for a high-risk population in eastern China[6]. The authors analyzed eight variable selection techniques—four regression-based and four tree-based—across two clinical datasets. Regression methods excelled in smaller datasets, while tree-based approaches, like RF, performed better in larger ones. This emphasized the effect of dataset size on the method's efficacy for clinical predictive modeling[7]. The authors emphasize the importance of selecting and reducing variables when developing prediction models. They explore techniques, like elimination, forward selection, stepwise selection and all possible subset selection. They discuss stopping rules such as p values, AIC, BIC and Mallows Cp for selection. Their research underscores the need to choose the variables and methodologies to enhance the accuracy and efficiency of prediction models. With 352,660 participants in the UK Biobank cohort, they examine how reflecting a genetic risk score (PRS) for artery disease (CAD) changes the performance of existing pooled cohort equations for risk prediction. The findings indicate that adding PRS only slightly enhances accuracy for new CAD cases, boosting C statistics from 0.76 to 0.78 and enhancing reclassification, particularly for certain individuals. However, the authors of these findings state that more investigation is needed to determine the impact of adding genetic information to CAD risk assessments although these findings are very important. They propose a machine learning approach to improve disease prediction by identifying important features for better accuracy[8]. While experimenting with different features and methods for classification, a team developed a model called HRFLM which combines a linear model and a random forest. This model picked out heart disease with an accuracy of 88.7%, showing how new, more sophisticated machine learning approaches can be useful in diagnosis of disease for the improvement of healthcare. The research article explains how machine learning (ML) helps in healthcare by explaining the automatic analysis of electrocardiograms (ECGs). It presents ML's ability to enhance the sensitivity and speed of detecting anomalies and do so early on, thus leading to better patient outcomes. One of the most important advantages of ML is that it can discover patterns and structures in the data that may be missed by other methods which can result in better diagnostic and treatment decisions[9,10]. However, the article also mentions the limitations of ML in the form of biases and interpretability issues that must be overcome to fully harness its potential in healthcare. The study involved using algorithms on data from 2829 patients to differentiate between individuals without issues and those with heart problems achieving an accuracy rate close to 85%[11]. It employed a network guided by principal component analysis data augmentation for classification purposes and network analysis to detect patterns, in heart variability. This study effectively distinguished between individuals, with health and those with conditions like fibrillation or congestive heart failure[12–14]. The research introduces the Modified Artificial Plant Optimization (MAPO) algorithm for identification of coronary heart disease (CHD) by analyzing heart rates captured in fingertip videos. MAPO accurately forecasts heart rate significantly reducing the complexity of datasets (by 81.25%) for examination. When combined with machine learning algorithms MAPO improves the accuracy of predicting CHD presence surpassing optimization techniques. This approach presents an avenue for diagnosis and prevention of CHD[14]. The article delves into the impact of AI on medicine highlighting its role in imaging, risk assessment and drug development. By discussing AIs progress and challenges such as data privacy and biases it underscores AIs potential to transform the treatment and understanding of heart disease[15]. The study investigates the use of machine learning (ML) and heart rate variability (HRV) to anticipate risk in patients with hypertension. It identifies HRV characteristics over a day, evaluates them thoroughly and selects the predictors. A model that combines decision trees and boosting achieved an accuracy rate of 97.08% indicating improvements in predicting events, among hypertensive patients[16]. The researchers in the study explored how artificial intelligence can help detect stress by testing models using heart rate data and evaluating their ability to learn from contexts. They found that the Local Outlier Factor and Multi-layer Perceptron models performed well in knowledge transfer management, especially apps[17]. They were able to detect anomaly too and create training models in those various suggested scenarios. combining A datasets

system to improve developed stress through detection a in device wearable and stress app to technology predict and disease machine (CVD) learning risk algorithms. levels This system[18] an achieved combination a F1 of great scores Internet potential for of for risk Things early categories, (IoT) assessment and of CVD it risks, has a stacking classifier with advanced machine learning algorithms improved its effectiveness as a CVD risk assessment tool. A network and other methods such as random forests and Bayesian networks were used to develop a model to predict the probability of CVD based on a dataset of 70,000 patients. The accuracy of the model presented confirms the potential of the application of machine learning methods for the improvement of the diagnosis and treatment of cardiovascular diseases. An AI model was also developed based on data from 1888 patients in care units to predict mortality rates. This model identified crucial predictors such as the Glasgow Coma Scale and heart rate variability, and it performed as accurately as the SAPS 2 system. This research article also described the application of ML and DL methods for identifying heart disease from signals using real heart sound datasets. A feature ensemble method and data augmentation technique were used to improve the dataset, and a multilayer perceptron model achieved an accuracy rate of 95.65%. detecting, these diagnosing diverse and research assessing studies diseases from (CVD). These field advancements illustrate the working development with datasets ML and of and additional walking AI predictors speed in and self-reported health status. Furthermore, applying ML techniques to analyze heart audio signals for disease identification is also presented. Decision, trees, Random Forests (RF), neural networks and Anomaly detection models have been used in the analysis of health data. Many studies have improved the accuracy of CVD prediction compared to models, which means that a potential example, ML uses for predicting CVD risks through devices or based on these in algorithms technologies incorporating such can these as be innovative the applied technologies Modified into Artificial practice real-world Plant for settings[19–22].

The choice of LR, RF, GBM, and MLP was based on their respective advantages for structured clinical data. LR functions as a clear baseline model, which provides excellent feature relationship interpretation in clinical settings. The RF and GBM algorithms provide ensemble-based robustness and perform well in high-dimensional, tabular health data, making them suitable for real-world deployments because they are interpretable and require minimal preprocessing. The deep learning spectrum MLP was selected to detect possible nonlinear patterns in patient risk profiles. Our selected models outperform traditional GBM while offering better accuracy and computational efficiency than more recent models like XGBoost, which only marginally outperforms traditional GBM but require additional fine-tuning. The application of CNNs to purely tabular CVD datasets is limited by their complexity, demand for large datasets, and lack of interpretability, an essential criterion for clinical decision support systems. The selected models proved effective while matching the requirements of clinical interpretability and resource-constrained environments.

Materials and methods

This is illustrated by Fig. 1 that describes a basic data analysis process of selecting a raw dataset and transforming it into usable knowledge using a logical data analysis workflow. The raw data is shown as a purple box which contains the initial unprocessed information. This workflow is then taken through a critical stage of preprocessing contained within a green box which gives an account of a set of primary steps that are to be taken to enhance the quality of the dataset. They include data cleaning to correct faulty or inconsistent data, distribution analysis to identify the data's probability distribution and tendencies, correlation analysis to identify the presence of statistical relationships among variables, and visual analysis to provide graphical representation of data characteristics. After these very detailed preprocessing steps, the workflow moves to the tools—building last and stage tuning represented off by predictive a model. gear This with stage consists of using machine learning techniques and statistical techniques on the cleaned data to develop models that can explain variables or respond. The presence of directional arrows shows the sequence of the workflow; this highlights the importance of data cleaning before any analytical modeling can commence. This visual representation also highlights the significance of proper preprocessing in improving the quality of subsequent analytical models.

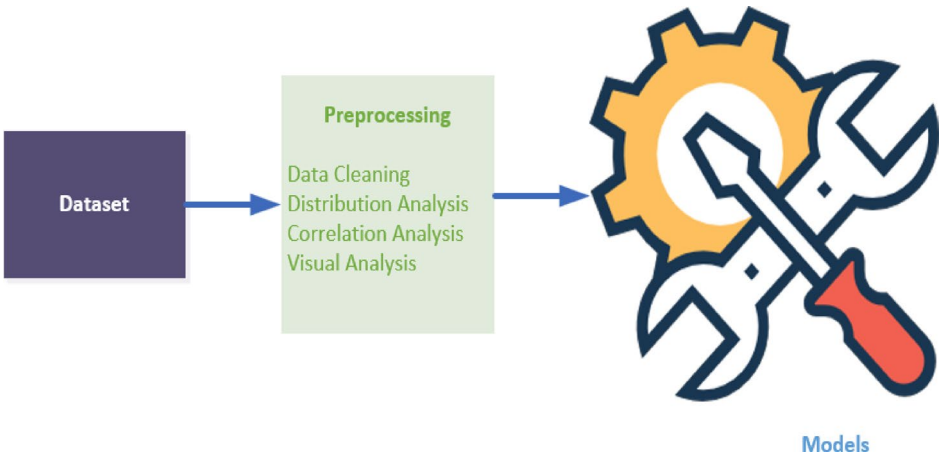


Fig. 1. Workflow from Raw Data to Predictive Modeling, with Preprocessing as the Key Intermediate Stage.

To accelerate our analysis, we leveraged a massive dataset of 70,000 entries (the dataset used from the Kaggle (https://www.kaggle.com/datasets/sulianova/cardiovascular-disease-dataset). Each entity served as a window into an individual's cardiovascular health, apprehending details like age, height, weight, blood pressure, cholesterol, smoking habits, alcohol intake, and physical activity. Before diving in, we thoroughly cleaned and prepped the data. This involved tackling missing values and eliminating outliers, ensuring a foundation of integrity and reliability. Furthermore, we employed feature engineering to extract even more meaningful insights.

In this study, a publicly available Kaggle dataset consisting of 70,000 samples has been used but it is not without some limitations. For instance, possible imbalances in demographics may restrict the transferability of our findings to other populations. Furthermore, the initial exploration analysis of the data revealed some problems with its quality, for example, unrealistic blood pressure values and possible omission of some factors in lifestyle variables (smoking, alcohol consumption etc.). However, many of these were addressed through extensive preprocessing; however, residual biases may still be present. It is therefore important that these models are tested in future studies on external datasets that are more diverse to validate their clinical applicability and reliability.

Exploratory data analysis

- *Balanced Target*: The target variable (likely indicating presence/absence of cardiovascular disease) is well-balanced, which is a positive aspect for training machine learning models.
- *Numerical Features*: Blood pressure readings require attention due to outliers and implausible values. Age, height, and weight distributions are generally good, though weight shows a slight rightward skew (meaning more data points on the higher end).
- *Categorical Features*: Cholesterol and glucose levels are strong indicators of cardiovascular disease in this dataset. While physical activity appears to be protective, the effects of smoking and alcohol require further investigation.
- *Cleaning Actions*: Outliers, particularly in blood pressure, need to be addressed before proceeding with model building.

It also helps to avoid the skewing of results by outliers, which are extremely high or low values that can unduly affect analysis. Finally, visualizing your data before and after cleaning can also be helpful to confirm if the removal of outliers was proper and efficient. Histograms or box plots are very plot types that can be very useful in demonstrating how your data distribution has been transformed by your cleaning process.

The distributions of the numerical features provide several insights

For a detailed visualization of the numerical feature distributions, please refer to Fig. 2.

- *Age*: The dataset leans towards younger patients, with a larger concentration of participants between 40 and 60 years old. This is reflected in the left-skewed distribution.
- *Height*: Both genders seem well-represented with a normal height distribution. Most individuals fall within the 150–180 cm range.
- *Weight*: Weight distribution skews slightly to the right, meaning there are more participants in the 50–100 kg range. A few outliers exist on the higher end, requiring further investigation.
- *Blood Pressure*: While both systolic (ap_hi) and diastolic (ap_lo) blood pressure distributions show a concentration around healthy ranges (100–140 mmHg systolic and 60–90 mmHg diastolic), there are concerned outliers. These include negative values, extremely high values, and potentially inaccurate data points that need to be addressed before analysis.

This analysis also considered the distribution of categorical and binary features, such as smoking status, alcohol consumption, and physical activity levels. However, for detailed insights into the distribution of these features, please refer to Fig. 3.

Model Selection and Deployment: A range of ML and DL models were selected for this study to compare their effectiveness in predicting cardiovascular diseases. The ML models included:

- Logistic Regression (LR): A statistical model used for binary classification tasks, which predicts the probability of occurrence of an event by fitting data to a logistic function.
- Random Forest (RF): An ensemble learning method that operates by constructing a multitude of decision trees at training time and outputting the class that is the mode of the classes (classification) of the individual trees.
- Gradient Boosting Machines (GBM): A machine learning technique that builds predictive models in a stage-wise fashion like other boosting methods, but it generalizes them by allowing optimization of an arbitrary differentiable loss function.

In the realm of deep learning, we utilized:

- Multilayer Perceptron (MLP): A feedforward artificial neural network (ANN) that consists of at least three layers of nodes: an input layer, a hidden layer, and an output layer. MLPs use a technique called backpropagation for training.

Evaluation Metrics: To assess the performance of each model, we employ several statistical and machine learning metrics:

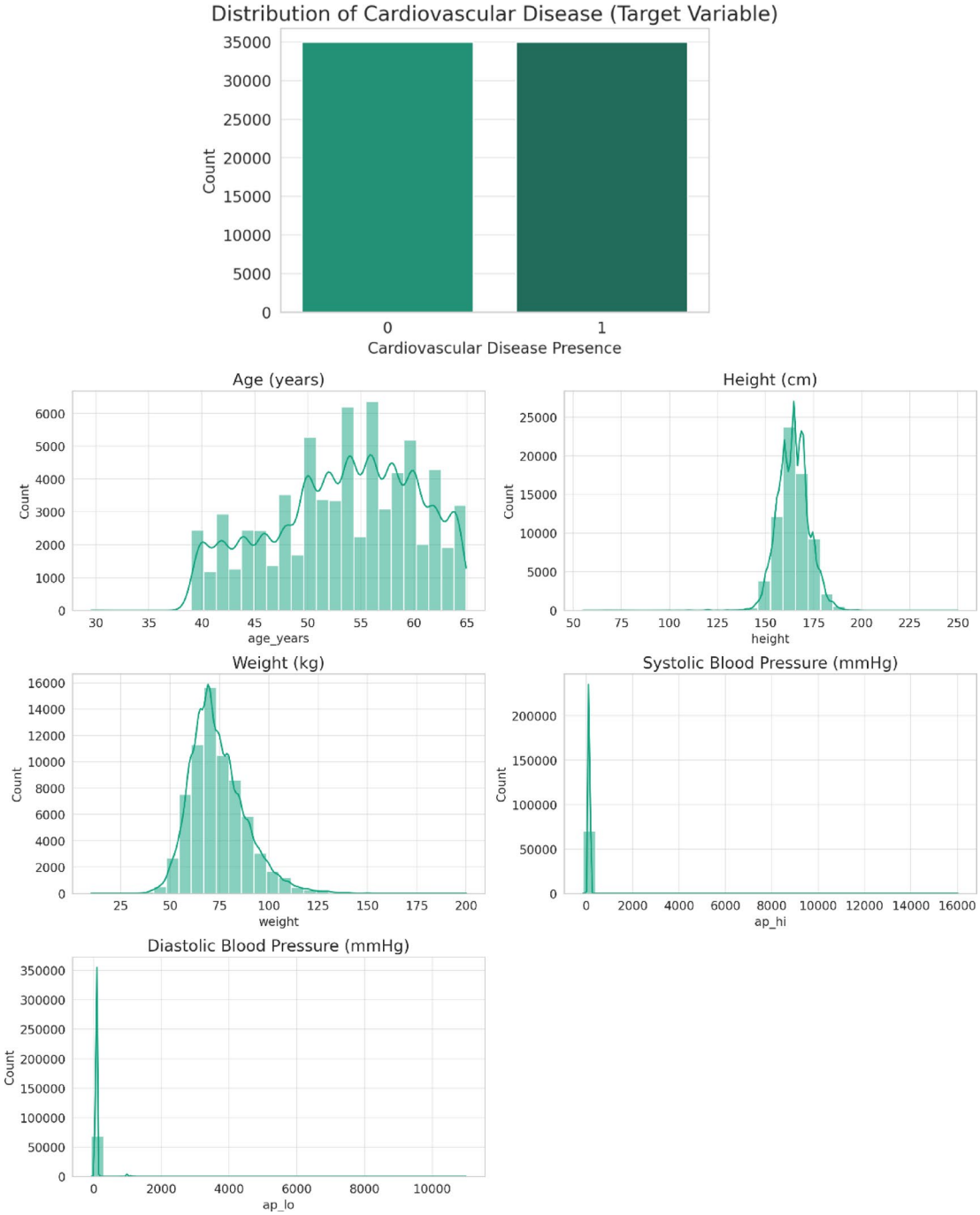


Fig. 2. Exploratory data analysis of the dataset.

- Accuracy: The ratio of correctly predicted observations to the total observations.
- Precision: The ratio of correctly predicted positive observations to the total predicted positives.
- Recall (Sensitivity): The ratio of correctly predicted positive observations to all observations in the actual class.
- F1 Score: The weighted average of Precision and Recall. This score takes both false positives and false negatives into account.
- ROC-AUC (Receiver Operating Characteristic - Area Under Curve): A comprehensive evaluation metric for binary classification models that measures the trade-off between sensitivity and specificity (true positive rate vs. false positive rate) at various threshold settings.

Statistical Analysis: All models were trained on a randomly selected 80% split of the dataset. The remaining 20% (10% is also used for validation purpose) was held back as a test set to evaluate model performance. Standard statistical tests, including chi-squared for independence and t-tests for mean comparisons, were applied to

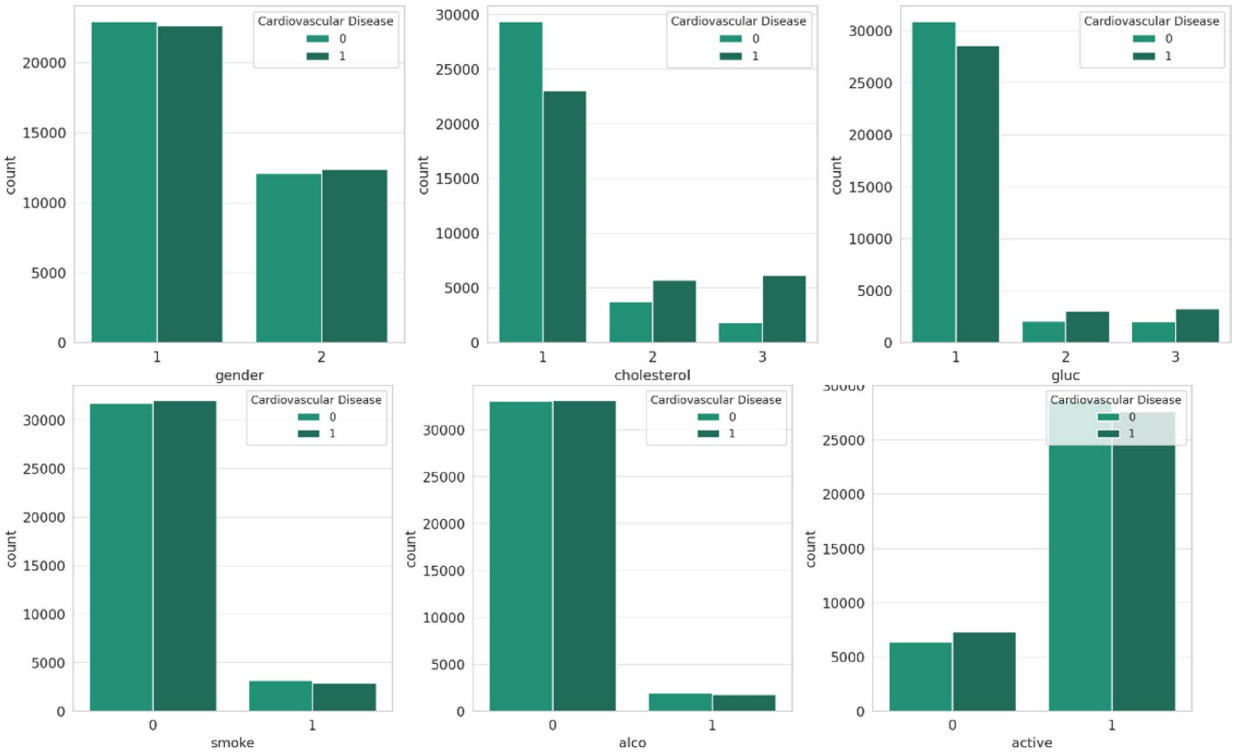


Fig. 3. Distribution of features.

explore the relationships between features and cardiovascular disease outcomes. Model training and testing were conducted using a ten-fold cross-validation approach to ensure the robustness and generalizability of the models.

A heatmap of the correlation matrix for the cardiovascular dataset was generated (Fig. 4), providing valuable insights into the relationships between different variables.

- Age: It displayed a moderate positive correlation with the target variable "cardio" (0.24), suggesting older individuals had a higher prevalence of cardiovascular disease.
- Weight and Height: Their correlations with cardiovascular outcomes were low to moderate (0.18 for weight and − 0.01 for height). This indicates that while weight had some influence, height did not significantly affect cardiovascular disease risk.
- Systolic (ap_hi) and Diastolic Blood Pressure (ap_lo): They showed strong correlations with the cardiovascular disease outcome (0.43 and 0.33 respectively), highlighting them as significant risk factors.
- Cholesterol and Glucose Levels: Both displayed positive correlations with cardiovascular outcomes (0.22 for cholesterol and 0.09 for glucose), underlining their importance in predicting cardiovascular health.

Lifestyle factors, including smoking, alcohol consumption, and physical activity, exhibited weaker correlations with cardiovascular disease compared to physiological measures. Physical activity, however, displayed a negative correlation (-0.04), suggesting a slight protective effect. Analysis of the correlation matrix (Fig. 2) revealed blood pressure readings (both systolic and diastolic) as the most significant predictors of cardiovascular disease in this dataset. Cholesterol and age also emerged as important factors, highlighting the need to prioritize lipid management and consider age-related risks in cardiovascular health strategies. The weak correlations observed for smoking and alcohol consumption could be attributed to underreporting or a potentially lower impact of these factors compared to directly measured physiological data like blood pressure and cholesterol. These findings can serve as valuable guidance for feature selection in predictive modeling. The emphasis should be placed on blood pressure, cholesterol, and age as key variables for building robust prediction models.

Results and discussion

Logistic regression

This study explored the use of logistic regression (LR) to predict cardiovascular disease (CVD) using a comprehensive dataset encompassing demographics, health indicators, and lifestyle choices (refer to Table 1 for detailed variable breakdown). Following meticulous cleaning and preparation, the data provided a strong foundation for modeling. The initial LR model yielded valuable insights into CVD risk factors, achieving an accuracy of 73%, a precision of 75%, and a recall of 67% (refer to Table 1 for a full breakdown of performance metrics). This balance between identifying true positive cases and overall accuracy is promising. The model's ability to differentiate between healthy individuals and those at risk was further confirmed by a strong initial

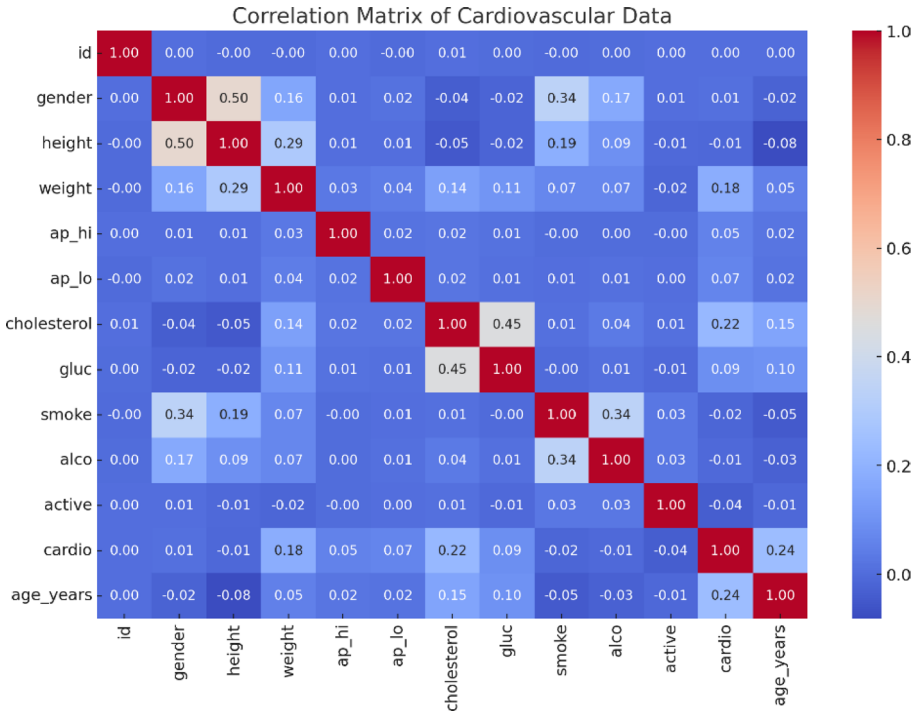


Fig. 4. Correlation matrix of various features of the dataset.

LR Performance on Various settings						
without Hyper tuning	Accuracy	Precision	Recall	F1	ROC-AUC	Hyperparameters
	72.82%	75%	67%	71%	73%	None
With Hyper tuning	73%	73%	73%	73%	79%	C approximately 0.17 and penalty of l2

Table 1. Overall result of LR under varying conditions.

Optimization Technique	Application
Regularization	L1 Regularization (Lasso) and L2 Regularization (Ridge):
Solver Choice	Liblinear and Saga
Hyperparameter Tuning	GridSearchCV (adjusting C, selecting penalty, and choosing the solver)

Table 2. Optimization techniques.

ROC-AUC score of 0.73. Hyperparameter tuning and cross-validation techniques (details provided in separate Tables 1 and 2, and 3) further optimized the model. While accuracy remained relatively stable, the ROC-AUC score significantly increased to 0.79, indicating enhanced performance and generalizability. The model's consistent F1 score around 0.71, even after optimization, highlights the robustness of LR for CVD prediction under varying conditions.

Hyperparameter tuning significantly improved the model's diagnostic ability. While accuracy, precision, and F1-score saw slight increases, the most notable improvement was in the ROC-AUC score, which jumped from 0.73 to 0.7951. This increase in ROC-AUC signifies a more robust ability to differentiate between healthy individuals and those with cardiovascular disease (CVD), making the model a more reliable prediction tool. Feature ranking (details in Table 4) revealed the relative importance of each variable. Systolic blood pressure (ap_hi) emerged as the most influential predictor, aligning with established knowledge of its strong association with heart disease and stroke. This emphasizes the critical role of strict blood pressure management for patients. Age (age_years) as a significant factor highlights the progressive nature of CVD and the cumulative impact of other risk factors over time. This finding supports the implementation of age-specific screening and intervention strategies. The significant coefficient for cholesterol levels underscores the importance of lipid management in preventing CVD. This may lead to more proactive screening and management for patients with borderline or high cholesterol. While its coefficient is smaller, the significance of BMI points towards the role of obesity in cardiovascular health. This suggests the potential benefit of incorporating weight management interventions into

Items	Conclusion
Cross-Validation Score: Average 5-Fold CV Score: 0.7268	This score suggests that the model, on average, correctly predicts outcomes with about 72.68% accuracy across different subsets of the training data. Cross-validation helps ensure that the model's performance is robust and not overly fitted to the specific partition of the data used in training.
Test Set Evaluation: Accuracy: 0.7321	The accuracy on the test set is similar to the cross-validation score, which confirms consistency between the model's performance in training and testing. It correctly predicts whether an individual has cardiovascular disease 73.21% of the time.
ROC-AUC Score: ROC-AUC: 0.7960	The ROC-AUC score is a comprehensive measure that considers all possible classification thresholds. A score of 0.7960 is quite good, indicating that the model has a strong ability to distinguish between the two classes (patients with and without cardiovascular disease).
Confusion Matrix: True Negatives (TN): 5505 False Positives (FP): 1440 False Negatives (FN): 2217 True Positives (TP): 4491	This suggests that the model is better at identifying true negatives but still has room to improve, particularly in reducing the number of false negatives and false positives.
Classification Report Precision, Recall, and F1-Score: Class 0 (No CVD): Precision: 0.713 Recall: 0.793 F1-Score: 0.751 Class 1 (Yes CVD): Precision: 0.757 Recall: 0.670 F1-Score: 0.711	The metrics show decent performance, with slightly better precision for predicting positive classes (CVD presence). However, improving recall for the positive class could be a target for further model tuning, as it indicates the model misses a significant number of actual positive cases (lower sensitivity).

Table 3. Overall results and conclusion on each item after running hyperparameters on LR algorithms.

Feature Importance	Coefficient	Classification Metrics	Confusion Matrix and Classification Report	ROC-AUC	Accuracy
ap_hi	1.006372	Precision for No CVD: 71% Recall for No CVD: 79% Precision for Yes CVD: 76% Recall for Yes CVD: 67%	Confusion Matrix: (TN = 5494) compared to true positives (TP = 4498), (FN = 2210),	0.7951	73.19%
age_years	0.337424				
cholesterol	0.300132				
active	-0.095906				

Table 4. Feature engineering results.

CVD prevention programs. The negative coefficient for physical activity states again its protective role against CVD. Encouraging patients to maintain or increase their physical activity levels can be a crucial component of prevention programs. The model's accuracy (73.19%) and ROC-AUC score reflect it being effective at distinguishing between CVD and non-CVD cases. However, its true clinical utility also depends on the balance between sensitivity (recall) and specificity, along with the disease prevalence in the target population. About one third of actual CVD cases are missing by the model's recall of 67%, which may lead to under-treatment. This may require additional safeguards, such as more frequent follow-up of at-risk patients or the use of other diagnostic tests in a clinical setting.

Random forest

Using default settings, an RF classifier was used to predict cardiovascular disease (CVD) in a dataset; the model's performance was evaluated. The accuracy of the model was 71.41%, precision was 72% and recall was 70%. This indicates that the performance is well balanced. The F1-score of 71% represents a good balance of correctly identifying true positive cases (precision) and not identifying false negatives (recall). The model's ability to distinguish between people with and without CVD is also evident from the ROC-AUC score of 0.77 (Fig. 5). These results demonstrate the potential of the Random Forest model, even without hyperparameter tuning, as a robust tool for clinical support. It can assist healthcare providers in identifying patients at risk for CVD. Furthermore, key variables like systolic blood pressure and cholesterol levels emerged as significant predictors, aligning with established clinical knowledge regarding CVD risk factors (details in Table 5).

While the model achieved a respectable accuracy of 73.60%, indicating its potential as a medical diagnostic tool, there's room for improvement. This could involve exploring additional feature engineering techniques or alternative modeling approaches. Precision, at 76.07%, is commendable in a medical setting, where minimizing false positives is crucial. However, achieving a good balance with recall is essential. Recall, at 68.55%, presents the most significant opportunity for improvement. Enhancing this metric would reduce false negatives, a critical factor in disease prediction where missing actual cases can have serious consequences. The F1-score of 72.12% reflects a good overall model, striking a balance between precision and recall. However, there's potential to refine the model to improve recall without significantly sacrificing precision. The strong ROC-AUC score of 80.00%

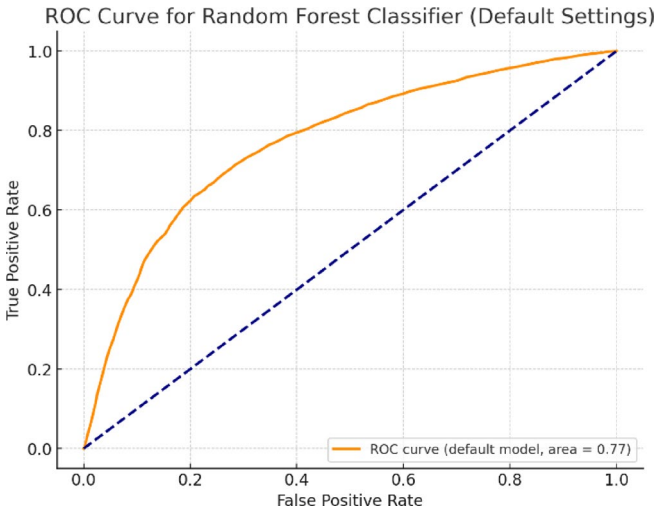


Fig. 5. ROC AUC curve before hyperparameter Tuning.

RF Performance on Various settings						
without Hyper tuning	Accuracy	Precision	Recall	F1	ROC-AUC	Hyperparameters
	71.41%	72%	70%	71%	77%	None
With Hyper tuning	73%	76%	68%	72%	80%	Best parameters found: {'bootstrap': True, 'max_depth': 10, 'max_features': 'sqrt', 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 100}

Table 5. Overall result of RF algorithm.

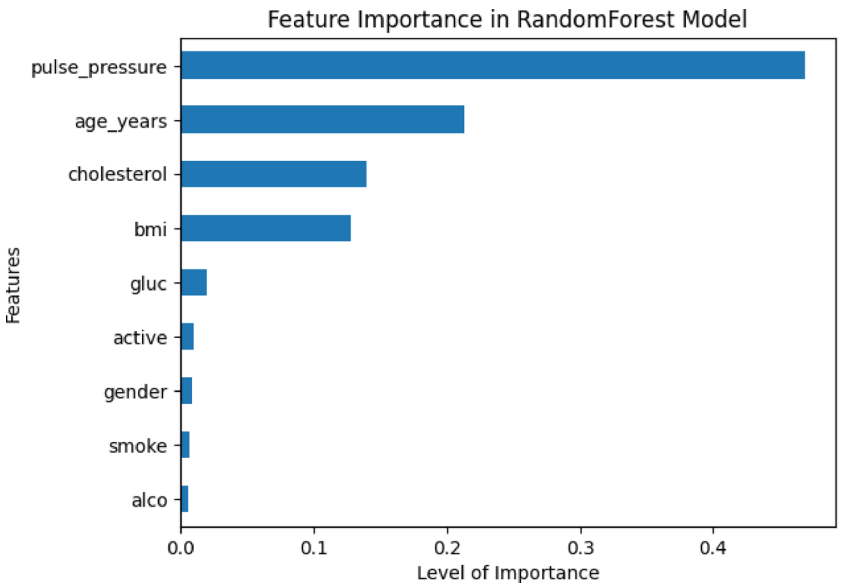


Fig. 6. Important feature RM model.

(refer to Fig. 6) indicates excellent discriminatory ability. This suggests the model effectively distinguishes between individuals with and without the disease across various thresholds.

A key takeaway from the random forest model's feature importance analysis (Fig. 7) is the critical role of physiological and biometric factors in predicting CVD.

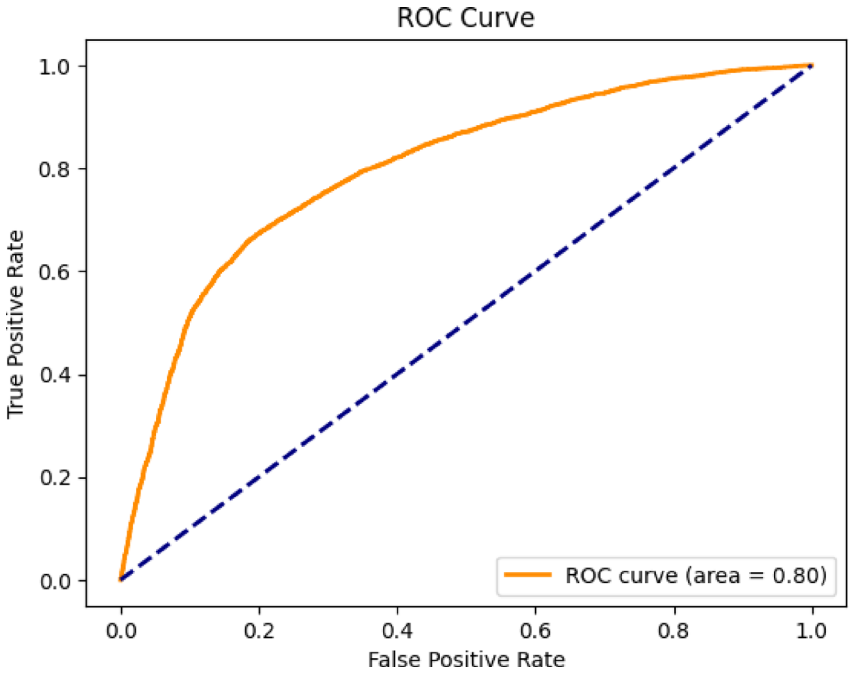


Fig. 7. ROC AUC curve after hyperparameter Tuning at {'bootstrap': True, 'max_depth': 10, 'max_features': 'sqrt', 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 100}.

Feature Importance	Coefficient	Classification Metrics	ROC-AUC
Pulse Pressure	0.468762	Precision: Class 0 (No CVD): 0.71 and Class 1 (Yes CVD): 0.77 Recall: Class 0 (No CVD): 0.79 and Class 1 (Yes CVD): 0.68 F1-Score: Class 0 (No CVD): 0.75 and Class 1 (Yes CVD): 0.72 Accuracy: Overall: 0.74	0.80
age_years	0.212723		
cholesterol	0.139966		
BMI	0.127714		

Table 6. Feature engineering results for RF model.

- Pulse pressure, representing the difference between systolic and diastolic blood pressure, emerges as the most influential factor. This aligns with clinical knowledge, as a large pulse pressure often indicates arterial stiffness and increased heart disease risk.
- Age follows closely, reflecting the well-established trend of rising CVD risk with age.
- Cholesterol and BMI also feature prominently, emphasizing their well-known connections to heart health. High cholesterol promotes plaque buildup, while a higher BMI is linked to obesity-related complications, both major contributors to heart disease.
- Interestingly, glucose levels have less influence compared to the others, despite their known role in cardiovascular complications, especially in diabetes.

Lifestyle factors like physical activity, gender, smoking, and alcohol consumption also play a part, but to a lesser extent. This suggests that while lifestyle choices influence cardiovascular health, the model prioritizes physiological and biometric features in its assessment. This analysis (Fig. 6) serves as a valuable guide for healthcare providers by highlighting the most critical factors in cardiovascular risk assessment. This can help them prioritize interventions and plan preventive measures effectively. However, it also emphasizes the need for a holistic approach considering both lifestyle and biological metrics. The model's performance metrics (Table 6) demonstrate promising accuracy in CVD identification. With an overall accuracy of 74% and a ROC-AUC score of 0.802, it effectively distinguishes between those with and without CVD.

- High precision (77%) suggests the model is correct in about three-quarters of its positive CVD predictions, crucial for minimizing false positives and unnecessary interventions.
- However, recall for positive cases (68%) indicates room for improvement. This means the model misses around 32% of actual CVD cases, potentially delaying crucial interventions with serious consequences.
- The model excels in identifying true negatives (no CVD) with a recall of 79%. This is important for ruling out patients who don't require further exams.

Future efforts to improve recall without losing precision could involve:

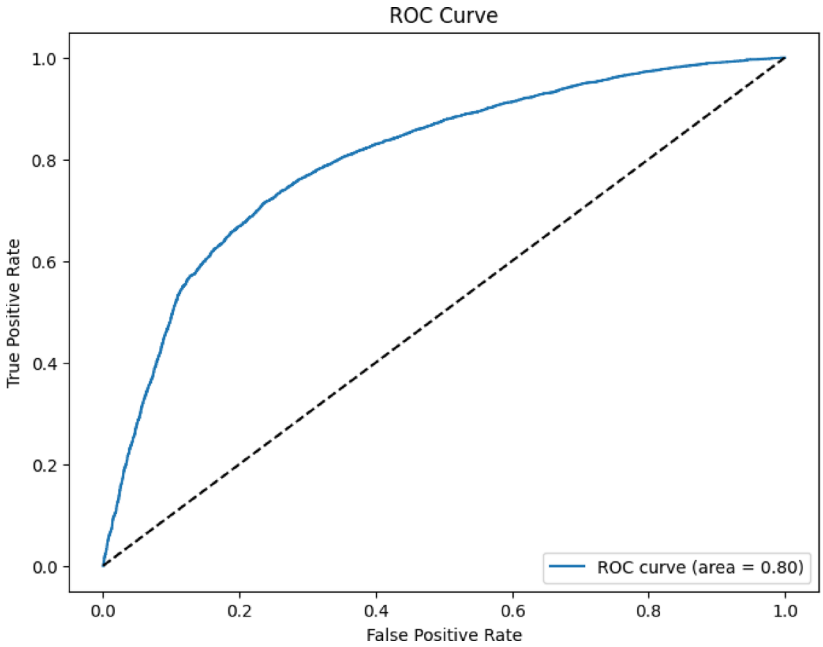


Fig. 8. ROC AUC of MLP.

GBM Performance on Various settings						
without Hyper tuning	Accuracy	Precision	Recall	F1	ROC-AUC	Hyperparameters
	73%	75%	70%	74%	80%	None
With Hyper tuning	74%	75%	71%	74%	80%	classifier__ learning_rate': 0.1, 'classifier__ max_depth': 3, 'classifier__max_ features': None, 'classifier__min_ samples_leaf': 2, 'classifier__min_ samples_split': 2, 'classifier__n_ estimators': 100}

Table 7. Overall result of GBM.

- Adjusting the decision threshold to capture more true positive cases.
- Advanced feature engineering and exploring complex ensemble methods to refine the model.

Enhancing recall would make the model even more valuable for clinical decision-making. The high ROC-AUC score further supports its applicability in differentiating between disease states with confidence. Overall, this model presents a promising tool for early CVD detection and diagnosis, supporting clinical decisions with machine learning accuracy. With further tuning, it has the potential to significantly reduce missed diagnoses and improve patient outcomes for cardiovascular health.

GBM

The Gradient Boosting model demonstrates robust performance in predicting CVD (Fig. 8). Cross-validation ROC-AUC scores consistently range from 0.795 to 0.805 with an average of 0.800. This signifies the model's consistent accuracy across different data splits, a crucial factor in real-world use. High ROC-AUC scores suggest the model effectively ranks patients based on their likelihood of having CVD, aiding in prioritizing clinical attention and resource allocation. On the test set (Table 7), the model achieves good precision for both classifying patients with and without CVD (72% and 75%, respectively). This indicates a slightly better ability to identify patients with the disease. Recall scores are 77% for negative cases (no CVD) and 70% for positive cases (CVD). While slightly better at identifying those without CVD, a recall of 70% for positive cases remains valuable. Missing true CVD cases can have severe consequences, so this level of recall is significant in a clinical setting.

Overall, the model exhibits a balanced performance:

- Accuracy of 73% on the test set suggests its general suitability for diagnostic support.

Feature Importance	Classification Metrics	ROC-AUC
'ap_hi'	Precision: Class 0 (No CVD): 72% and Class 1 (Yes CVD): 76% Recall: Class 0 (No CVD): 77%, and Class 1 (Yes CVD): 71% F1-Score: Class 0 (No CVD): 0.75 and Class 1 (Yes CVD): 0.73 Accuracy: Overall: 74	0.80
'ap_lo'		
cholesterol		
age		

Table 8. Overview of REF feature selection of the GBM.

Multilayer Perceptron Performance on Various settings						
without Hyper tuning	Accuracy	Precision	Recall	F1	ROC-AUC	Hyperparameters
	74	76%	78%	75%	80%	None
With Hyper tuning	74%	75%	77%	75%	80%	Best parameters found: {'activation': 'tanh', 'alpha': 0.0001, 'hidden_layer_sizes': (50,), 'learning_rate': 'constant', 'solver': 'adam'}

Table 9. Overall result of MLP model with and without hyperparameter tuning.

- Combined F1-scores of 74% and 72% for the two classes reflect a good balance between precision and recall.

A notable ROC-AUC score of 0.801 on the test set further strengthens the model's ability to differentiate between CVD and non-CVD cases. This score suggests a high-quality predictive model, particularly in the challenging domain of medical diagnosis. Overall, the Gradient Boosting model shows a promising tool for clinical decision-making in cardiovascular health.

The GBM establishes substantial effectiveness in predicting CVD. Assessments with and without hyperparameter tuning divulge consistent strengths. Both configurations achieve high accuracy (around 73%) and ROC-AUC scores (above 0.80), point to excellent ability to distinguish between CVD and non-CVD cases. F1-scores for both classes hover around 0.73–0.75, reflecting a good balance between precision (correctly identifying true positives/negatives) and recall (not missing true cases). This is crucial in clinical settings to avoid misdiagnoses. Hyperparameter tuning offers slight advantages such as improved sensitivity, it shows a slight improvement in identifying true positive CVD cases (recall of 71% vs. 70% without tuning). This can be crucial for catching more cases and potentially improving patient outcomes. Maintained Specificity: Precision for both CVD and non-CVD cases remains high (around 72–77%) with tuning, ensuring the model doesn't generate excessive false positives.

Feature selection using REF identified key predictors:

Age.

Systolic and diastolic blood pressure (ap_hi, ap_lo).

Cholesterol.

Age (years).

This aligns with established knowledge about CVD risk factors, highlighting the model's grounding in medical evidence. The RFE-selected GBM achieved an accuracy of 74% and a strong ROC-AUC score (0.7975), as shown in the Table 8. This suggests its effectiveness for early screening of CVD. Precision and recall for non-CVD cases: 72% and 77% respectively, indicating good proficiency in identifying healthy individuals. Precision and recall for CVD cases: 76% and 71% respectively. While the model identifies true positives with reasonable accuracy, there's room for improvement in capturing more cases. The balanced F1-scores and consistent macro/weighted averages across metrics demonstrate the model's overall effectiveness in various clinical scenarios. In conclusion, the GBM model, especially with hyperparameter tuning and RFE-selected features, offers a valuable tool for healthcare professionals. It can aid in identifying individuals at risk for CVD, potentially leading to earlier interventions and improved patient outcomes. Continuous monitoring and adaptation based on new data will be crucial for maintaining its effectiveness in real-world settings.

Multilayer perceptron (MLP)

Evaluating the Multilayer Perceptron (MLP) models for CVD diagnosis reveals promising results (Table 9). Both the basic and hyperparameter-tuned models achieve high ROC-AUC scores around 0.801, indicating a strong ability to differentiate between patients with and without CVD. Hyperparameter tuning offers a slight benefit, improving recall for CVD cases from 69 to 71% (Table 9). This suggests the tuned model is better at catching true positives. However, overall performance metrics like accuracy, precision, and F1-scores remain comparable across both models for the two classes (CVD and non-CVD) (Table 9). This highlights the effectiveness of the basic model even without extensive tuning. Overall, both basic and tuned MLP models show potential as valuable tools for clinical settings, aiding in accurate and timely CVD screening. However, ongoing refinement and validation against a broader range of clinical data are recommended to ensure continued reliable performance in diverse healthcare environments.

The permutation importance analysis of the MLP model for CVD prediction offers valuable clues regarding the most influential features. This information can be instrumental in refining the model and engineering even more effective features in the future.

Systolic blood pressure (ap_hi) reigns supreme, with an importance of around 0.144, significantly influencing the model's decisions. This aligns perfectly with medical knowledge, as high systolic blood pressure is a well-known CVD risk factor. Cholesterol (cholesterol) follows closely at 0.0379, reflecting its established connection to heart health. The model clearly recognizes both these factors as crucial for CVD prediction. Age (age_years) also plays a significant role (0.0219), with older individuals typically at higher risk. The model appropriately assigns weight to age as a predictive factor. Diastolic blood pressure (ap_lo) holds some influence (0.0067) although less than systolic pressure, highlighting its relevance in CVD assessments. Weight is likely an indicator of BMI or body fat levels showing an impact (0.0052) because it is connected to various risk factors, for heart disease. Similarly, maintaining a lifestyle with a significance of 0.0036 highlights the influence of choices on heart health. Factors such as height, glucose levels, gender and smoking are considered crucial based on their importance scores in predicting cardiovascular disease outcomes within this model. It is worth noting that alcohol consumption has slightly negative importance (0.00003) suggesting that it may not significantly contribute to predictions or could have a relationship with other variables, in the dataset. These findings reinforce the importance of traditional cardiovascular risk factors like blood pressure, cholesterol, and age in disease prediction models. The analysis can guide model refinement by focusing on the most impactful features. This could lead to a simpler, faster, and less prone to overfitting model without sacrificing accuracy. For instance, reducing emphasis or even removing features with minimal impact could streamline the model.

The revised MLP model, focusing on a select group of key features (age, systolic blood pressure, cholesterol, weight, and activity level), exhibits promising results in diagnosing cardiovascular disease (CVD). This streamlined model maintains consistent performance across different data splits, with ROC-AUC scores ranging from 0.790 to 0.805 (average 0.798), as shown in Fig. 8. Its ability to differentiate between CVD and non-CVD patients is further confirmed by a strong ROC-AUC score of 0.8001 on the test set. Overall accuracy sits at around 73.74%, indicating a reliable prediction in nearly three quarters of cases. While both precision and recall metrics are positive, a slight trend emerges. The model is slightly better at identifying true positives for CVD (75%) compared to non-CVD (73%). However, recall scores reveal a stronger performance in correctly identifying non-CVD cases (76%) versus CVD cases (71%). The indication here is that the model may overlook cases of disease, which warrants additional scrutiny. However, the rounded F1 scores (0.74, for non-CVD and 0.73 for CVD) indicate an equilibrium between precision and recall underscoring its overall effectiveness. This equilibrium holds importance in environments to prevent incorrect diagnoses. In general, the updated MLP model displays potential as a resource for initial cardiovascular disease screening. Its impressive performance and accuracy can assist healthcare professionals in prioritizing patients for assessment and treatment planning.

Table 10 gives a summary of the factors that impact the risk of disease (CVD). Biometric information such as age, blood pressure (both systolic and diastolic) and cholesterol levels are known risk factors. While weight and height are mentioned, Body Mass Index (BMI) is considered a measure for assessing health risks. Lifestyle choices also play a role in CVD risk. High glucose levels indicating diabetes and smoking are factors while physical activity is beneficial. Interestingly gender may also have an impact on CVD risk. The study used Recursive Feature Elimination (RFE) to identify the features for predicting CVD. This analysis found that age blood pressure readings and cholesterol levels were the factors. This simplified the model. Highlighted predictors of CVD making it easier to comprehend.

The logistic regression model performed effectively in predicting CVD by achieving a separation between individuals and those at risk while maintaining precision (correctly identifying true positives) and recall (not missing actual cases). Adjusting model settings through hyperparameter tuning resulted in a ROC AUC score underscoring its importance, in enhancing model performance. The Importance of Focusing on Key Aspects; The emphasis on information by RFE highlights the significance of prioritizing these elements over lifestyle choices, which had a lesser impact on the models' forecasts. This indicates that accessible medical data can be reliable in pinpointing individuals at a heightened risk of disease. The Evolution of Cardiovascular Disease Forecasting; This research sets the stage for incorporating machine learning into healthcare environments. These algorithms can aid in detection and treatment strategies by pinpointing high risk patients based on essential medical data. Subsequent studies could expand upon this by delving into models and potential connections, among features leading to a deeper comprehension of how various factors affect cardiovascular disease risk.

Traditional ML models including Random Forest and Gradient Boosting are more interpretable, easier to explain and validate clinically, and require fewer computational resources, thus being more scalable and practical for real-world, resource-constrained clinical settings. Their predictive performance is moderately accurate to accurate, suiting the modest requirement for accuracy in diagnostic decisions. The interpretable nature of these models enhances learning and ensures transparency and confidence in their clinical application. On the other hand, advanced deep learning models like CNNs offer higher accuracy by capturing more complex and fine-grained features from the input data. However, their complexity and lack of interpretability limit their

Feature engineering	Basic Biometric Data	Lifestyle Indicators
df['bmi'] = df['weight'] / ((df['height'] / 100) ** 2) df['age_years'] = df['age'] / 365.25 df['pulse_pressure'] = df['ap_hi'] - df['ap_lo'] RFE: Index(['age_years', 'bmi', 'cholesterol', 'ap_hi', 'active'], dtype='object')	Systolic blood pressure (ap_hi) Diastolic blood pressure (ap_lo) Cholesterol BMI (Body Mass Index)	Glucose (gluc) Smoking (smoke), Alcohol intake (alco), and Physical activity (active)

Table 10. Comparing with clinical settings.

application in medicine, where transparency and explainability are crucial for patient care and decision-making. The decision to use traditional or deep learning models depends on the specific needs and priorities of the diagnostic task at hand: traditional models offer balance, accuracy, and interpretability, while deep learning models focus on achieving the highest accuracy, even if the price is reduced interpretability.

The predictors that our models identified as significant—especially age, systolic and diastolic blood pressure, cholesterol, and BMI—have a high degree of conformity with the traditional cardiovascular risk factors listed by the American Heart Association (AHA). These findings substantiate current clinical practice guidelines that focus on blood pressure and cholesterol control, weight regulation, and age-related screening in cardiovascular risk assessment. This consistency confirms our models' clinical validity, that they not only do well statistically but also produce predictions that are consistent with current best knowledge, thereby potentially enhancing the value and credibility of our models for clinical decision making.

Future work[23–25] could also greatly benefit from the consideration of more sophisticated feature selection strategies, including embedded strategies or DL-based feature selection to improve the accuracy and robustness of predictions. Moreover, the integration of complex ensemble strategies (e.g., stacking or voting classifiers that combine several traditional and DL models) may help to avoid overfitting and fully exploit the potential of different models. These directions may be useful for improving the understandability and the quality of the predictions, which is one of the major issues when applying clinical models. Such improvements may contribute to the improvement of the predictive ability, the generalization, and the clinical significance of the cardiovascular risk prediction models in real-world healthcare settings. The findings indicate that different hyperparameters were used under various situations and produced optimal results (Refer Table 11).

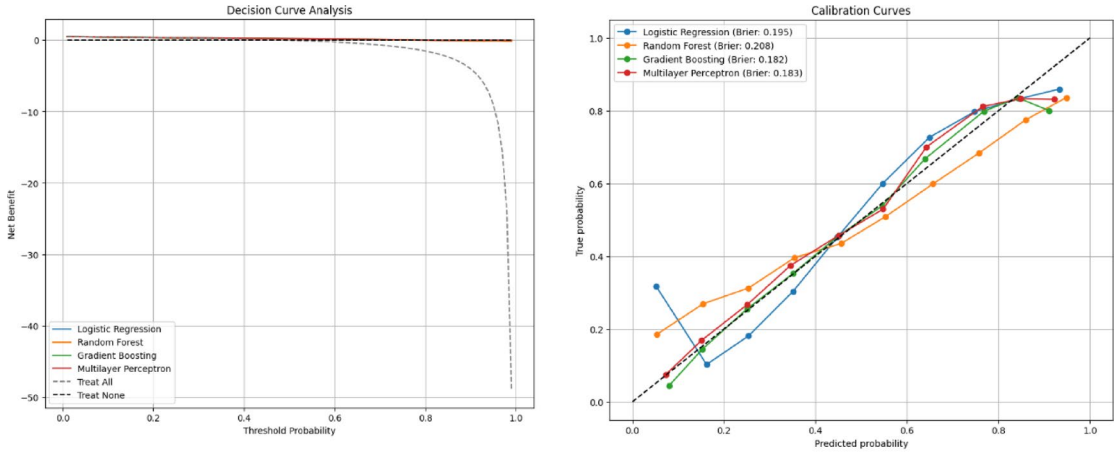
The clinical reliability of the models received additional evaluation through calibration analysis and decision curve analysis (DCA). The GB model (Brier score: 0.182) and MPP (0.183) demonstrated the best probability calibration through their close alignment with the ideal diagonal according to Fig. 9. The probability estimates of LR (0.195) and RF (0.208) showed moderate deviation from the ideal results. The Brier scores supported these findings by showing that lower values indicated better calibration. The decision curve analysis in Fig. 9 demonstrates the net benefit of each model across threshold probabilities ranging from 0.01 to 0.99. The four models demonstrated superior performance than both "Treat All" and "Treat None" strategies which indicate their potential to support clinical decision-making for intervention choices. Gradient Boosting and MLP demonstrated the highest net benefit throughout the clinically important threshold range of 0.1–0.3 which supports their application in risk-based decision-making frameworks for cardiovascular disease screening. Both models demonstrate calibrated probabilities and practical decision-support capabilities for real-world clinical use.

Conclusions

The research investigated how ML and DL models could work together to forecast CVD risk using a structured dataset. Advanced feature engineering together with interaction terms and recursive feature elimination techniques improved model performance and interpretability. The clinical environment benefits from LR and RF and GBM because these models deliver effective results while requiring low computational power and providing easy deployment capabilities. The MP DL model demonstrated superior performance in detecting

Model	Hyperparameter	Before Tuning (Default)	After Tuning (Optimized)
Logistic Regression	Regularization (C)	1.0 (default)	0.17
	Penalty	L2 (default)	L2
	Solver	lbfgs (default)	Liblinear, Saga
Random Forest	Bootstrap	True (default)	True
	Max Depth	None (default)	10
	Max Features	auto (default)	sqrt
	Min Samples Leaf	1 (default)	1
	Min Samples Split	2 (default)	2
	Number of Estimators	100 (default)	100
Gradient Boosting Machines (GBM)	Learning Rate	0.1 (default)	0.1
	Max Depth	3 (default)	3
	Max Features	None (default)	None
	Min Samples Leaf	1 (default)	2
	Min Samples Split	2 (default)	2
	Number of Estimators	100 (default)	100
Multilayer Perceptron (MLP)	Activation Function	relu (default)	tanh
	Alpha (Regularization)	0.0001 (default)	0.0001
	Hidden Layer Sizes	(100,) (default)	(50,)
	Learning Rate	constant (default)	constant
	Solver	adam (default)	adam

Table 11. Model and their various hyperparameter setting before and after turning them.



a B

Fig. 9. Calibration and decision curve analysis for four models. Gradient Boosting and MLP showed the best calibration (lowest Brier scores) and highest net clinical benefit across key thresholds.

complex nonlinear patterns but needed significant computational resources and produced less transparent results. Our results underscore that no single model is universally superior. The selection of models depends on specific circumstances because it requires a balance between predictive accuracy and interpretability together with operational feasibility. ML models are suitable for resource-limited or real-time applications, yet DL models provide better performance in scenarios with extensive high-dimensional data and adequate computational resources. The research provides a strong base for data-based CVD risk prediction while demonstrating the need for customized model approaches. Future studies need to incorporate various real-time data sources such as wearables and electronic health records to enhance personalized cardiovascular risk evaluations in clinical practice.

Data availability

The datasets generated during and/or analyzed during the current study are available in the Kaggle repository, [https://www.kaggle.com/datasets/sulianova/cardiovascular-disease-dataset /].

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Author contributions

G.I.A.J. and M.K. made substantial contributions to the conception and design of the work, the analysis and interpretation of the data, and the creation of new software used in the work. M.K. drafted the initial manuscript. Moreover, G.I.A.J. and M.K. substantively revised the manuscript. G.I.A.J. and M.K. approved the submitted version of the manuscript and any substantially modified versions that involve the author's contribution to the study. G.I.A.J. and M.K. agreed to be personally accountable for the author's own contributions and ensured that questions related to the accuracy or integrity of any part of the work, even ones in which the author was not personally involved, are appropriately investigated, resolved, and the resolution documented in the literature.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to G.I.A.J.

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